

GNU-Plot-Lab-Notebook

Roll Number: _____

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Semester: _____

Section 1

Question 1

Problem Set 1 (ii). Define $y = f(x) = x^2 \sin(x)$ and plot it for $-12 \leq x \leq 12$. Show both axes, label them `x values` and `my function`, and set the x -tick interval to 2.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3
4  set ou "Lab-files/Output/1/Q_ii_ou.svg"
5
6  set xrange [-12:12]
7
8  set xzeroaxis linetype 1 lc "black"
9  set yzeroaxis linetype 1 lc "black"
10
11 set xtics 2
12
13 set xlabel "x values"
14 set ylabel "my function"
15
16 set grid
17
18 y(x)=(x**2)*sin(x)
19
20
21
22 plot y(x) lc "#EE6677" lw 2
```

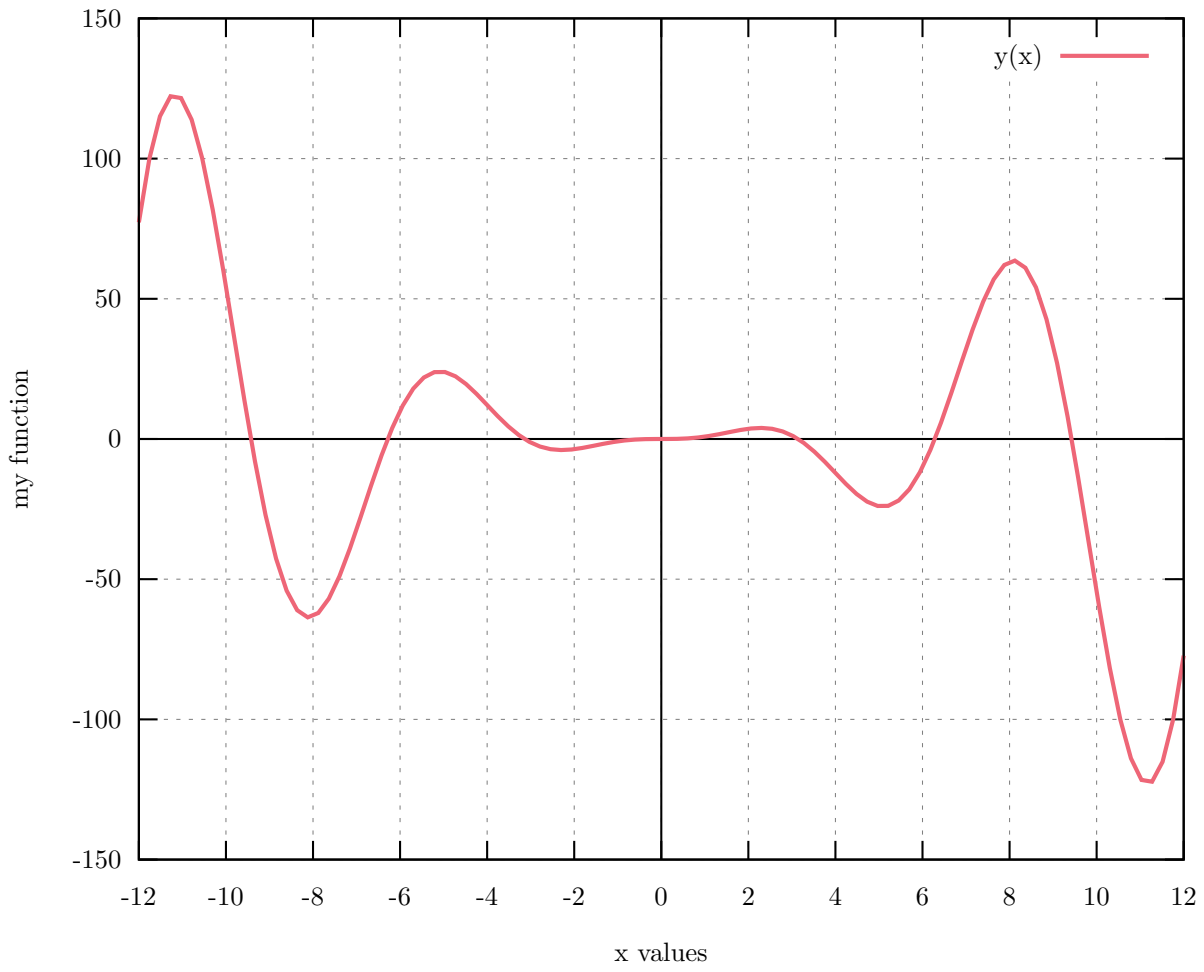


Figure 1: Output for Problem Set 1 (ii).

Question 2

Problem Set 1 (vii). Plot $f(x) = \cos(x) - 0.5x$ for $-20 \leq x \leq 20$ and $-10 \leq y \leq 12$. Show the x -axis, label the axes x value and $\cos(x) - 0.5x$, and set the x -tick interval to 2.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3
4  set ou "Lab-files/Output/1/Q_vii_ou.svg"
5
6
7  f(x)=cos(x)-(0.5)*x
8
9  set xrange [-20:20]
10 set yrange [-10:12]
11
12 set grid
13
14 set xzeroaxis linestyle 1 lc "black"
15 set yzeroaxis linestyle 1 lc "black"
16
17 set xtics 2
18
19 set xlabel "x value"
20 set ylabel "cos(x)-0.5x"
21
22
23 plot f(x) lw 2 lc "#4477AA"
```

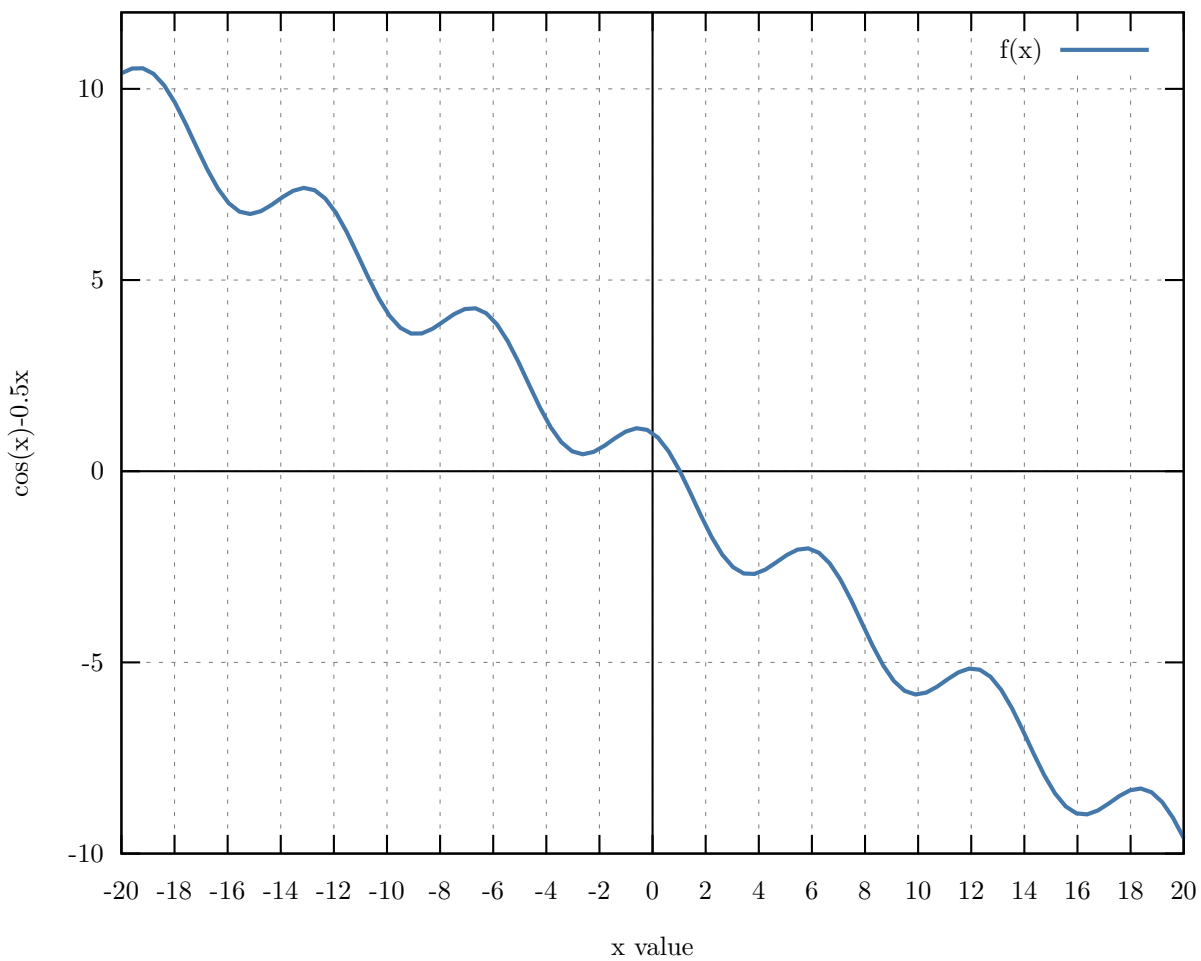


Figure 2: Output for Problem Set 1 (vii).

Question 3

Problem Set 1 (viii). Plot the piecewise function

$$f(x) = \begin{cases} x, & x \geq 2, \\ x^2, & -3 \leq x \leq 2, \\ -x, & x \leq -3 \end{cases}$$

for $-8 \leq x \leq 8$ and $-2 \leq y \leq 12$. Set the x -tick interval to 2 and label the axes **x value** and **jerk function**.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/1/Q_viii_ou.svg"
4
5  set grid
6
7  set xtics 2
8  set ytics 1
9
10 set xzeroaxis linestyle 3 lc "black"
11 set yzeroaxis linestyle 3 lc "black"
12
13 set xrange [-8:8]
14 set yrange [-2:12]
15
16 set xlabel "x value"
17 set ylabel "jerk function"
18
19 set samples 10000
20
21 f(x)=(x<=-3)?(-x):(x<=2)?(x**2):(x)
22
23 plot f(x) lc rgb "#332288" lw 2
```

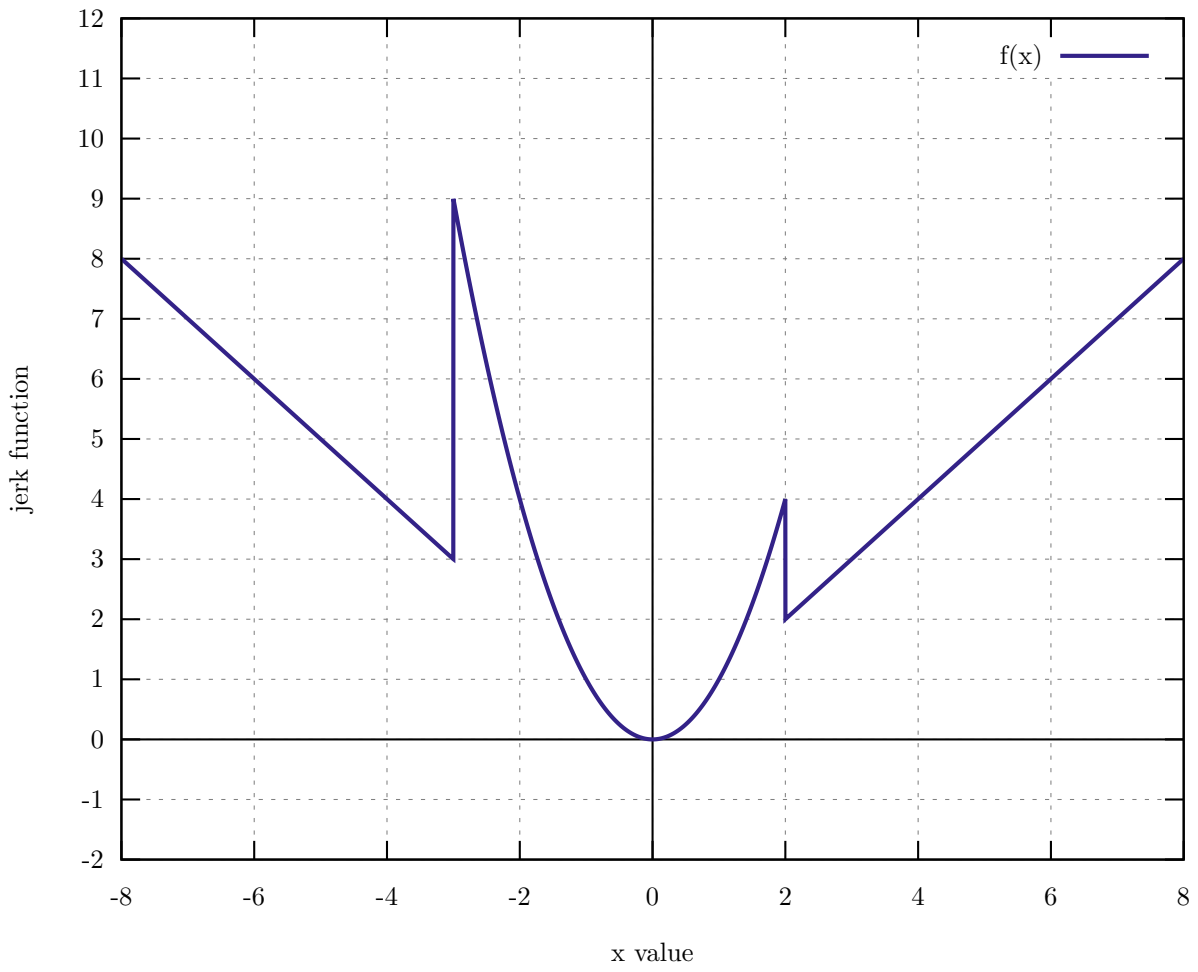


Figure 3: Output for Problem Set 1 (viii).

Question 4

Problem Set 1 (ix). Plot the functions

$$f_1(x) = x \ (x > 2), \quad f_2(x) = x^2 \ (-1 < x < 2), \quad f_3(x) = 2 \ (x < -1)$$

for $-6 \leq x \leq 6$ and $-2 \leq y \leq 7$. Set the x -tick interval to 1 and label the axes `x values` and `piecewise continuous functions`.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/1/Q_ix_ou.svg"
4
5  set grid
6
7  set xrange [-6:6]
8  set yrange [-2:7]
9
10 set xtics 1
11
12 set xlabel "x values"
13 set ylabel "piecewise continuous functions"
14
15 set xzeroaxis linestyle 1 lc "black"
16 set yzeroaxis linestyle 1 lc "black"
17
18
19 f3(x)=(x<-1)?(2):NaN #Not a Number
20 f2(x)=(x>-1 && x<2)?(x**2):NaN
21 f1(x)=(x>2)?(x):NaN
22
23
24 plot f1(x) lc "#332288" lw 2 ,f2(x) lc "#EE7733" lw 2 ,f3(x) lc "#44AA99" lw 2
```

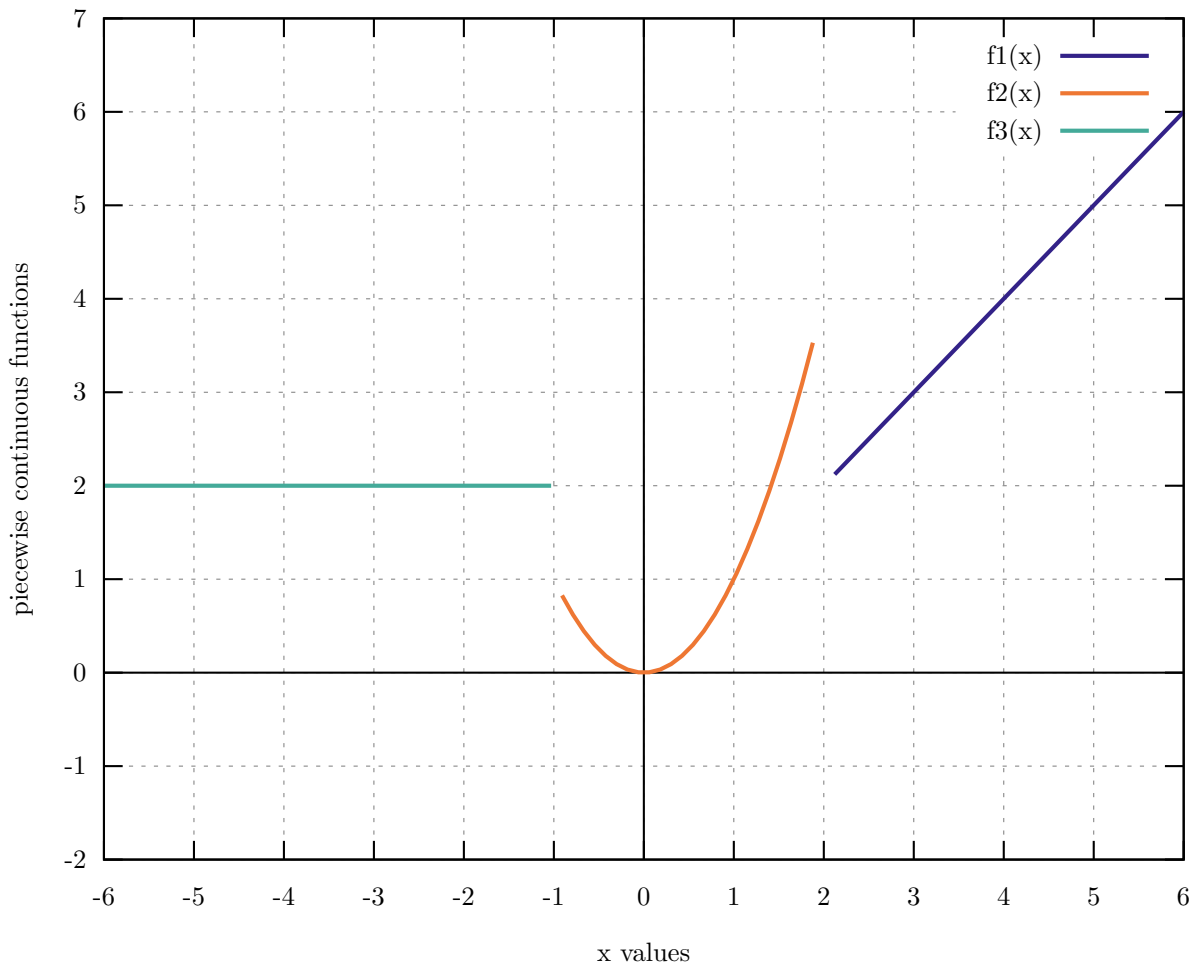


Figure 4: Output for Problem Set 1 (ix).

Question 5

Problem Set 1 (x). Plot $\sinh^2(x)$, $-\cosh(x)$, and $-(1+\tanh(x))$ together over $-6 \leq x \leq 6$ and $-8 \leq y \leq 8$. Use progressively increasing line widths and label the axes **x values** and **hyperbolic functions**.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/1/Q_x_ou.svg"
4
5  set grid
6
7  set xrange [-6:6]
8  set yrange [-8:8]
9
10 set xtics 1
11
12 set xlabel "x values"
13 set ylabel "hyperbolic functions"
14
15 set xzeroaxis linestyle 1 lc "black"
16 set yzeroaxis linestyle 1 lc "black"
17
18
19 f1(x)=(sinh(x))**2
20 f2(x)=-cosh(x)
21 f3(x)=-(1+tanh(x))
22
23
24
25 plot f1(x) lc "#44AA99" lw 1 ,f2(x) lc "#EE7733" lw 2 ,f3(x) lc "#4477AA" lw 3
```

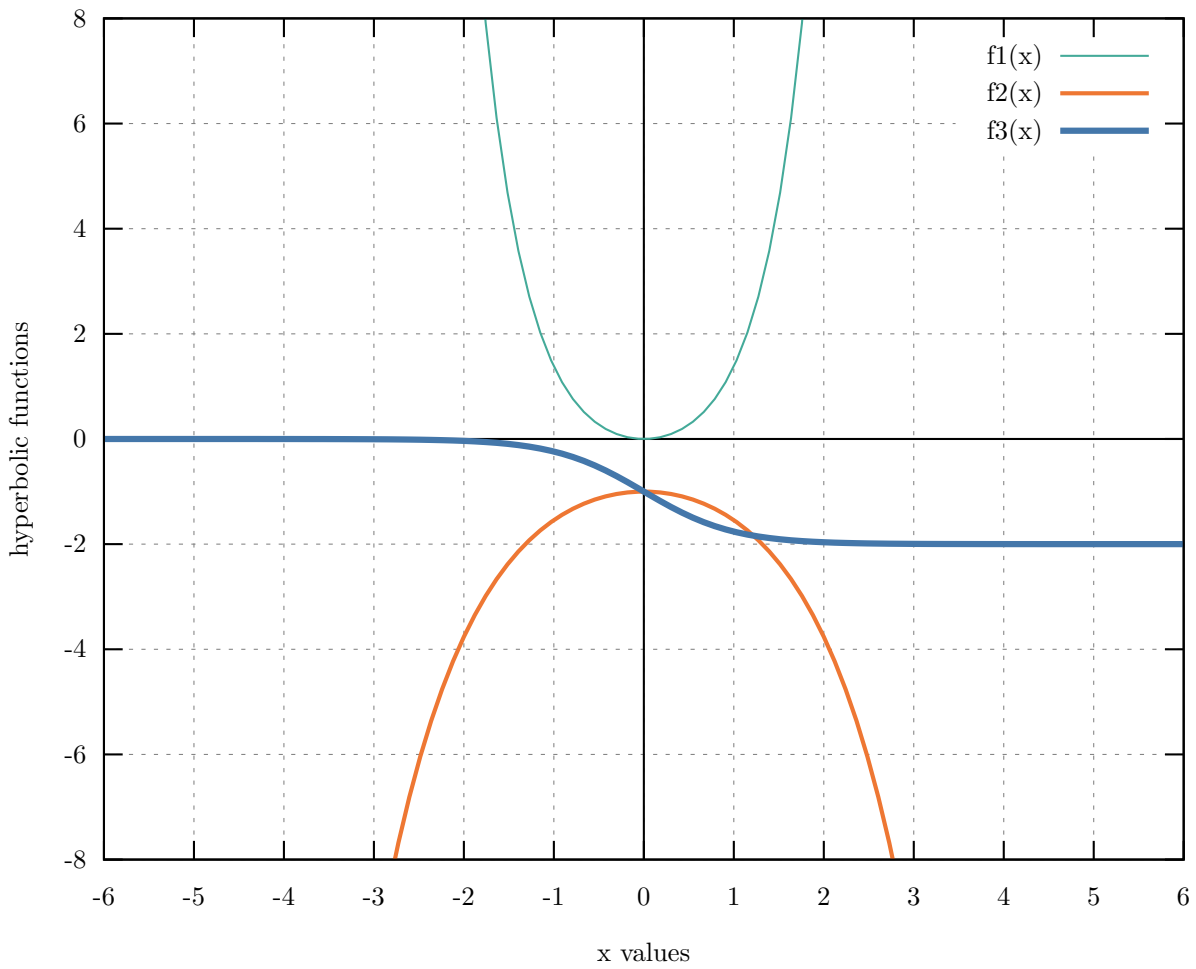


Figure 5: Output for Problem Set 1 (x).

Question 6

Problem Set 1 (xviii). Plot

$$f_1(x) = \frac{e^x}{x} \quad (x > 1), \quad f_2(x) = x^2 \tanh(x) \quad (-1 < x < 1), \quad f_3(x) = -\frac{e^{-x}}{x} \quad (x < -1)$$

for $-3 \leq x \leq 3$ and $-7 \leq y \leq 7$. Show both axes, label the y -axis Piecewise continuous function, and set the y -tick interval to 1.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set term svg background "white"
3  set ou "Lab-files/Output/1/Q_xviii_ou.svg"
4
5
6
7  set grid
8
9  set xzeroaxis linestyle 1 lc "black"
10 set yzeroaxis linestyle 1 lc "black"
11
12 set xrange [-3:3]
13 set yrange [-7:7]
14
15
16 set ytics 1
17
18 set ylabel "Piecewise continuous function"
19
20 f1(x) =(x>1)?((1/x)*exp(x)):NaN
21 f2(x) =(x>-1 && x<1)?((x**2)*tanh(x)):NaN
22 f3(x) =(x<-1)?((-1/x)*exp(-x)):NaN
23
24 plot f1(x) lw 2 lc "#4477AA" , f2(x) lw 2.5 lc "#44AA99" , f3(x) lw 2 lc "#EE7733"
```

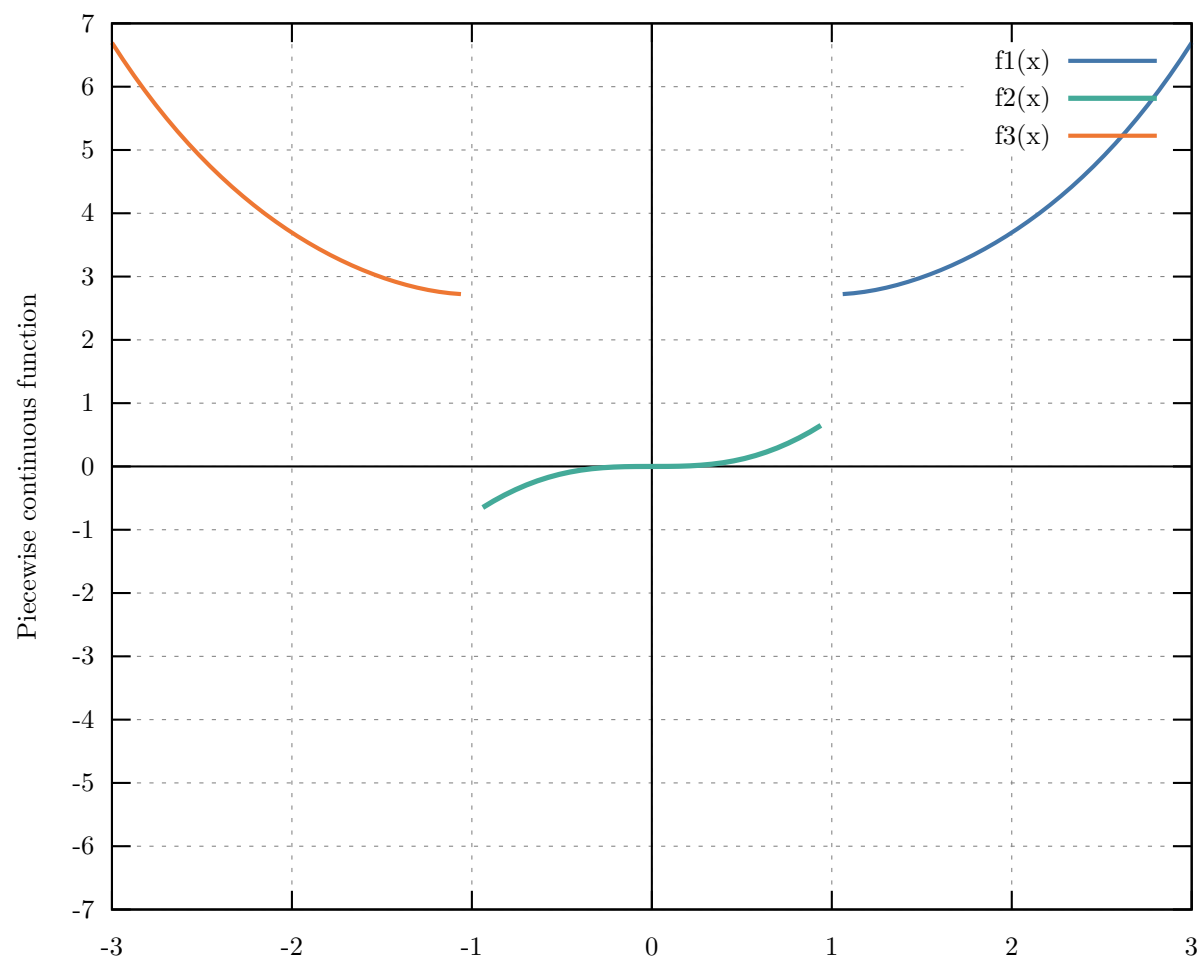


Figure 6: Output for Problem Set 1 (xviii).

Question 7

Problem Set 1 (xix). Plot the piecewise functions

$$f(x) = 1.45x^3 - 6\sqrt{x} \ (x \geq 2), \quad g(x) = \sin^2(x) \ (-1 \leq x < 2), \quad h(x) = |x|e^x \ (x < -1)$$

for $-2 \leq x \leq 3$ and $-5 \leq y \leq 5$. Set the x - and y -tick intervals to 1 and 2, respectively, and title the plot *Piece-wise continuous functions*.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set term svg background "white"
3  set ou "Lab-files/Output/1/Q_xix_ou.svg"
4
5
6
7  set title "Piece-wise continuous functions"
8
9  set grid
10
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14 set xrange [-2:3]
15 set yrange [-5:5]
16
17 set xtics 1
18 set ytics 2
19
20 f(x)=(x>=2)?(1.45*(x**3) - 6*(x**(0.5))):NaN
21 g(x)=(x>=-1 && x<2)?((sin(x))**2):NaN
22 h(x)=(x<-1)?(abs(x)*exp(x)):NaN
23
24
25 plot f(x) lw 2 lc "#EE7733" , g(x) lw 2 lc "#4477AA" , h(x) lw 2 lc "#228833"
```

Piece-wise continuous functions

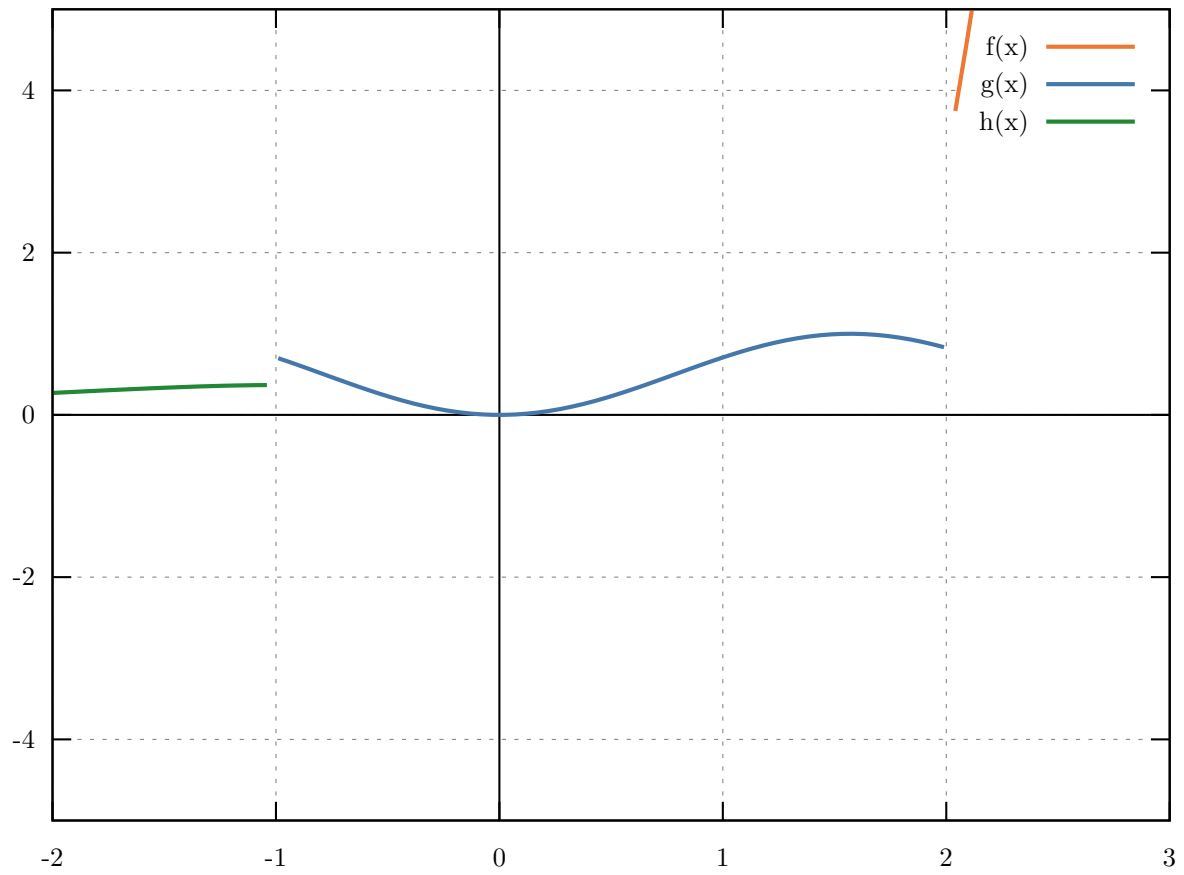


Figure 7: Output for Problem Set 1 (xix).

Section 2

Question 1

Problem Set 2 (i). Fit the data with $y = mx + c$ over $-4 \leq x \leq 8$. Plot the points and fitted line, choosing a y -range that allows the intercept c to be read from the y -axis. Report m and c .

Table 1: Data for Problem Set 2 (i).

x	y
-2.7	-15.07
-1.5	-9.5
1.4	3.72
3.3	12.5
6.1	25.2

Data file contents:

```
-2.7 -15.07
-1.5 -9.5
1.4 3.72
3.3 12.5
6.1 25.2
```

Sol: The fitted plot and parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_i_ou.svg"
4
5  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_i.log"
6
7  set grid
8  set xzeroaxis linestyle 1 lc "black"
9  set yzeroaxis linestyle 1 lc "black"
10
11 set xrange [-4:8]
12
13 # Fitting-curve
14
15 y(x)=m*x + c
16
17 # Initial guess
18 m=1
19 c=-1
20
21 #Fitting the curve
22 fit y(x) "Lab-files/Data/2/Q_i.txt" using 1:2 via m,c
23
24 #Plotting
25 plot "Lab-files/Data/2/Q_i.txt" using 1:2 with p ps 1.5 lc "#4477AA" title "Data
    points", \
26     y(x) lw 2 lc "#EE7733" title "Fitted line"
```

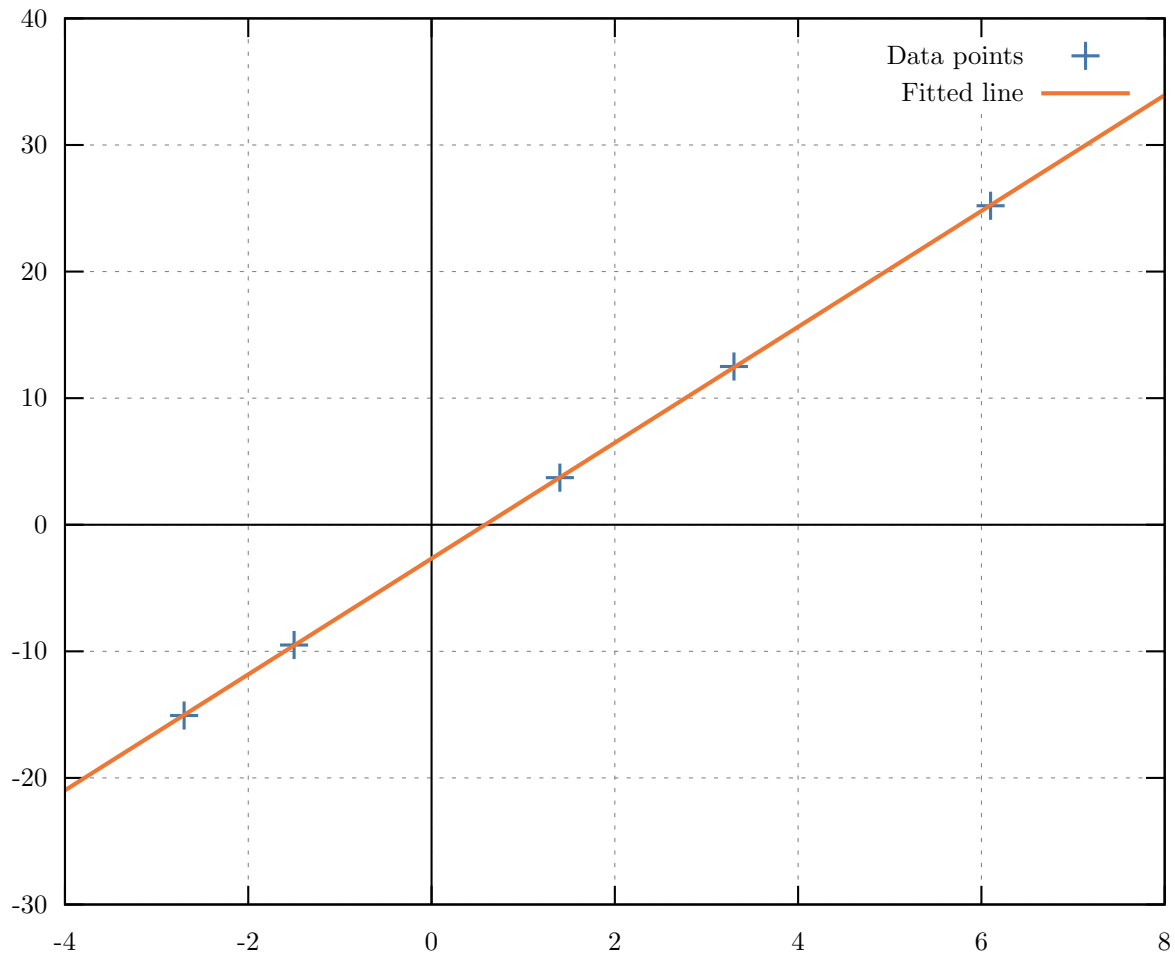


Figure 8: Linear fit for Problem Set 2 (i).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 4.57623	+/- 0.008141	(0.1779%)
c	= -2.67062	+/- 0.02811	(1.052%)

Question 2

Problem Set 2 (ii). Fit the data with $y = mx + c$ over $-7 \leq x \leq 7$. Plot the points and fitted line, show the y -axis, and choose a y -range that allows c to be read from it. Report m and c .

Table 2: Data for Problem Set 2 (ii).

x	y
-5.1	21.8
-3.4	16.12
-0.5	7.05
1.57	0.42
6.2	-14.2

Data file contents:

```
-5.1 21.8  
-3.4 16.12  
-0.5 7.05  
1.57 0.42  
6.2 -14.2
```

Sol: The fitted plot and parameter estimates with their standard errors are shown below.

```
1 reset  
2 set terminal svg background "white"  
3 set ou "Lab-files/Output/2/Q_2_ii_ou.svg"  
4  
5  
6 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_ii.log"  
7  
8  
9 set grid  
10 set xzeroaxis linestyle 0 lc "black"  
11 set yzeroaxis linestyle 0 lc "black"  
12  
13 set xrange [-7:7]  
14 #Fitting_curve  
15 y(x)=m * x + c  
16  
17 #Initial guess  
18 m=-1  
19 c=5  
20 #Curve fitting  
21 fit y(x) "Lab-files/Data/2/Q_ii.txt" using 1:2 via m,c  
22 #Plotting  
23 plot "Lab-files/Data/2/Q_ii.txt" using 1:2 with p ps 1.5 lc "#228833" title "Data  
    points", \  
24     y(x) lw 2 lc "#AA3377" title "Fitted line"
```

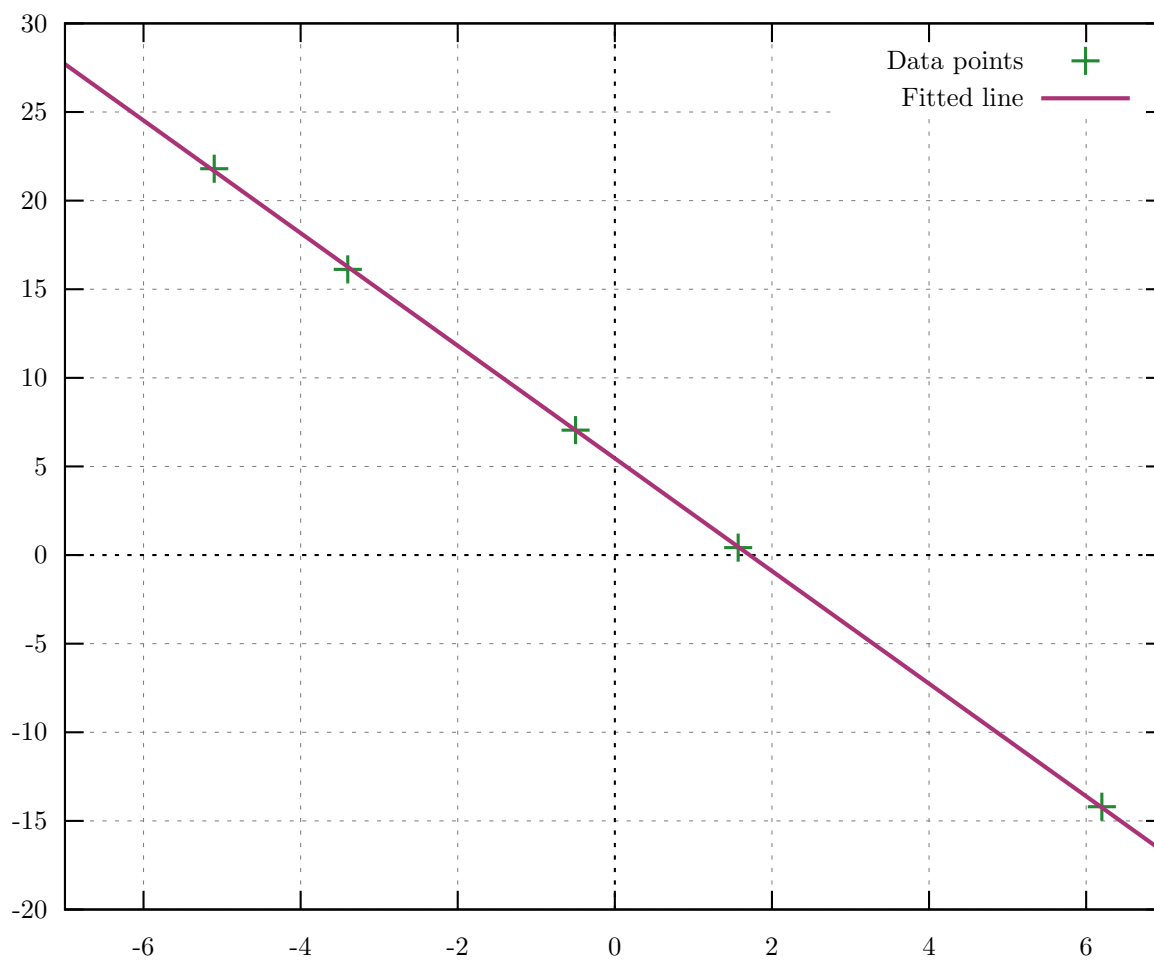


Figure 9: Linear fit for Problem Set 2 (ii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= -3.17793	+/- 0.01351	(0.4251%)
c	= 5.45623	+/- 0.05361	(0.9825%)

Question 3

Problem Set 2 (iv). Fit the data with $f(x) = mx + c$ over $-3 \leq x \leq 4$. Plot the points and fitted line, show both axes, and choose a y -range that allows c to be read from the y -axis. Report m and c .

Table 3: Data for Problem Set 2 (iv).

x	y
-2.0	-4.15
-1.3	-2.40
1.8	6.60
2.5	8.98
3.6	11.94

Data file contents:

```
-2.0 -4.15  
-1.3 -2.40  
1.8 6.60  
2.5 8.98  
3.6 11.94
```

Sol: The fitted plot and parameter estimates with their standard errors are shown below.

```
1  reset  
2  set terminal svg background "white"  
3  set ou "Lab-files/Output/2/Q_2_iv_ou.svg"  
4  
5  
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_iv.log"  
7  
8  set grid  
9  set xzeroaxis linestyle 0 lc "black"  
10 set yzeroaxis linestyle 0 lc "black"  
11  
12 set xrange [-3:4]  
13  
14 y(x)=m*x+c  
15 c=0.1  
16 m=1  
17  
18 fit y(x) "Lab-files/Data/2/Q_iv.txt" using 1:2 via m,c  
19  
20 plot "Lab-files/Data/2/Q_iv.txt" using 1:2 ps 1.5 lc "#CC6677" title "Data points"  
21 , \  
    y(x) lw 2 lc "#228833" title "Fitted line"
```

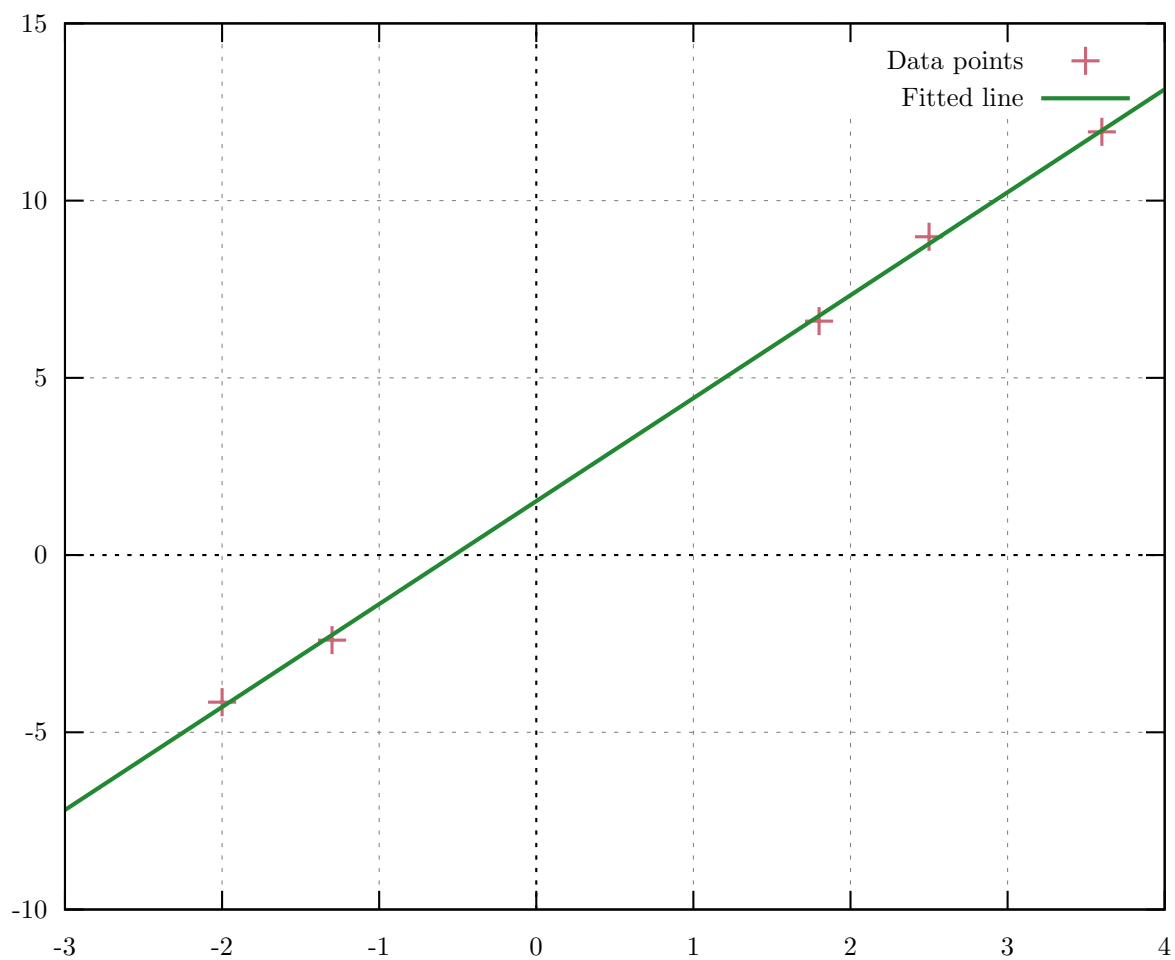


Figure 10: Linear fit for Problem Set 2 (iv).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 2.90453	+/- 0.03786	(1.304%)
c	= 1.52183	+/- 0.08983	(5.902%)

Question 4

Problem Set 2 (v). Fit the data with $f(x) = ax^b$ over $0 \leq x \leq 3$ and $0 \leq y \leq 35$. Plot the points and fitted curve, title it *Power series fit*, and report a and b .

Table 4: Data for Problem Set 2 (v).

x	y
0.5	0.21
0.83	1.05
1.16	2.40
1.82	9.01
2.81	30.75

Data file contents:

```
0.5 0.21
0.83 1.05
1.16 2.40
1.82 9.01
2.81 30.75
```

Sol: The fitted plot and parameter estimates with their standard errors are shown below.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_v_ou.svg"
4
5
6 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_v.log"
7
8 set title "Power series fit"
9 set grid
10 set xzeroaxis linestyle 0 lc "black"
11 set yzeroaxis linestyle 0 lc "black"
12
13 set xrange [0:3]
14 set yrange [0:35]
15
16 y(x)=a*(x**b)
17
18 a=1
19 b=1
20
21 fit y(x) "Lab-files/Data/2/Q_v.txt" using 1:2 via a,b
22
23 plot "Lab-files/Data/2/Q_v.txt" using 1:2 with p ps 1.5 lc "#EE7733" title "Data
    points", \
24     y(x) lw 2 lc "#4477AA" title "Fitted curve"
```

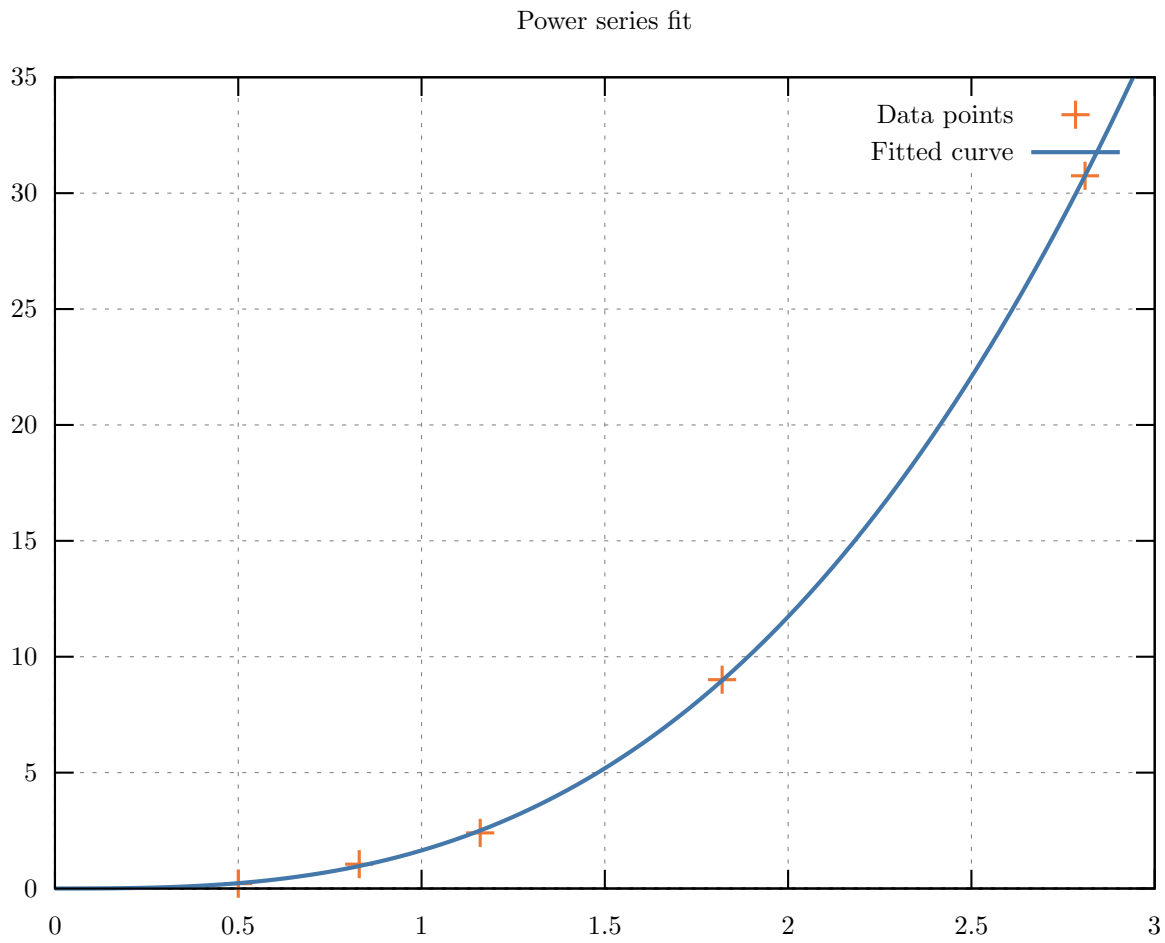


Figure 11: Power-series fit for Problem Set 2 (v).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 1.64269	+/- 0.02914	(1.774%)
b	= 2.8356	+/- 0.01767	(0.6232%)

Question 5

Problem Set 2 (vi). Fit the data with $f(x) = mx + c$ over $-3 \leq x \leq 3$. Plot the points and fitted line, show both axes, and choose a y -range that allows c to be read from the y -axis. Report m and c .

Table 5: Data for Problem Set 2 (vi).

x	y
-2.9	-5.96
-1.4	-3.81
0.8	-1.10
1.7	0.21
2.5	1.30

Data file contents:

```
-2.9 -5.96
-1.4 -3.81
0.8 -1.10
1.7 0.21
2.5 1.30
```

Sol: The fitted plot and parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_vi_ou.svg"
4
5
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_vi.log"
7
8  set grid
9  set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11
12 set xrange [-3:3]
13
14 y(x)=m*x+c
15
16 c=-2
17 m=1
18
19 fit y(x) "Lab-files/Data/2/Q_vi.txt" using 1:2 via m,c
20
21 plot "Lab-files/Data/2/Q_vi.txt" using 1:2 with p ps 1.5 lc "#228833" title "Data
    points", \
22     y(x) lw 2 lc "#EE7733" title "Fitted line"
```

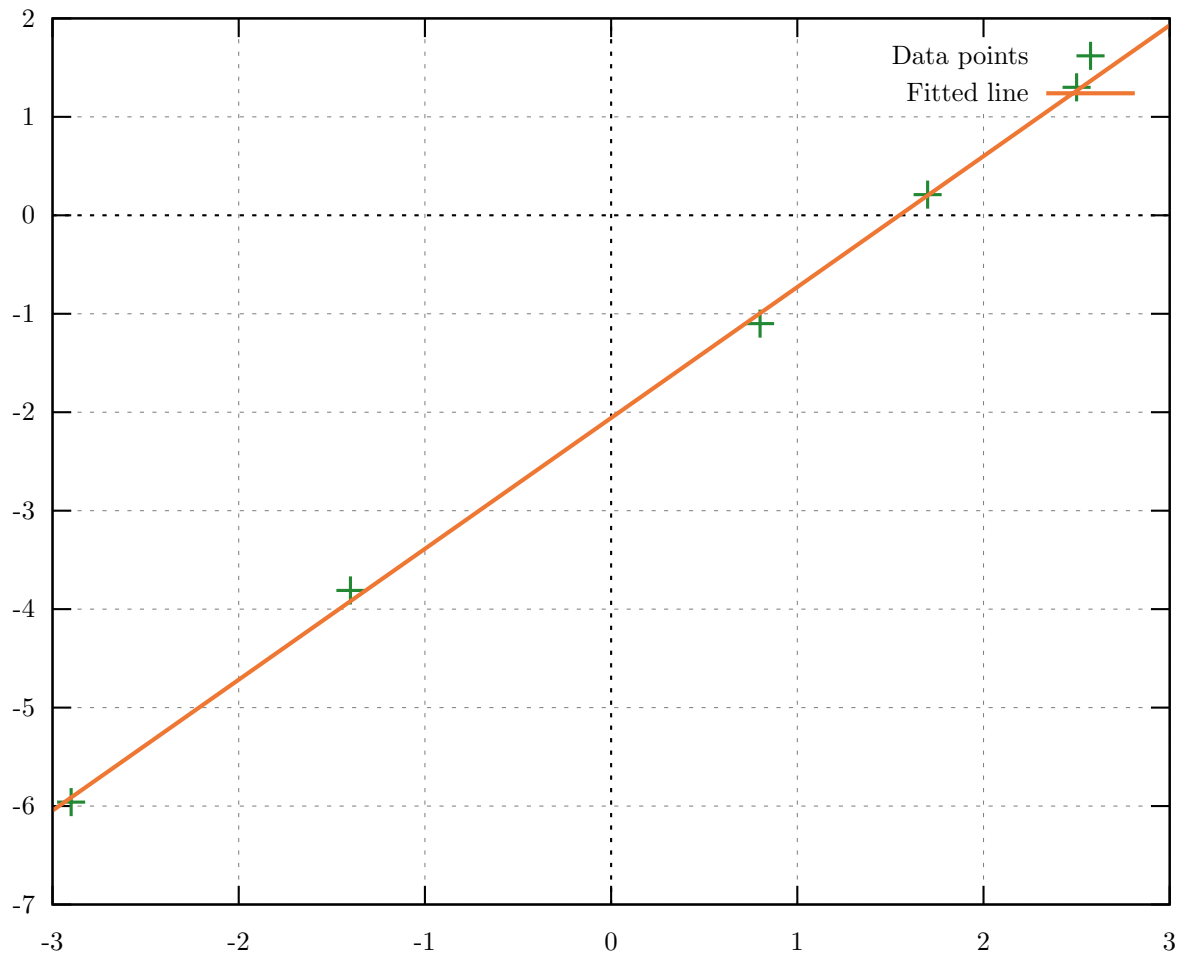


Figure 12: Linear fit for Problem Set 2 (vi).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 1.32931	+/- 0.02098	(1.579%)
c	= -2.0581	+/- 0.04213	(2.047%)

Question 6

Problem Set 2 (vii). Fit the data with $f(x) = ax^2 + b$ over $-3 \leq x \leq 3$ and $0 \leq y \leq 16$. Plot the points and fitted curve, title it *Polynomial fit*, and report a and b .

Table 6: Data for Problem Set 2 (vii).

x	y
-2.8	14.65
-1.2	5.81
0.7	4.48
1.8	8.33
2.9	15.56

Data file contents:

```
-2.8 14.65
-1.2 5.81
0.7 4.48
1.8 8.33
2.9 15.56
```

Sol: The fitted plot and parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_vii_ou.svg"
4
5  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_vii.log"
6  set title "Polynomial fit"
7  set grid
8  set xzeroaxis linestyle 0 lc "black"
9  set yzeroaxis linestyle 0 lc "black"
10
11 set xrange [-3:3]
12 set yrange [0:16]
13
14 y(x)=a*x**2+b
15
16 a=1
17 b=4
18
19 fit y(x) "Lab-files/Data/2/Q_vii.txt" using 1:2 via a,b
20 plot "Lab-files/Data/2/Q_vii.txt" using 1:2 with p ps 1.5 lc "#AA3377" title "Data
    points", \
21     y(x) lw 2 lc "#4477AA" title "Fitted curve"
```

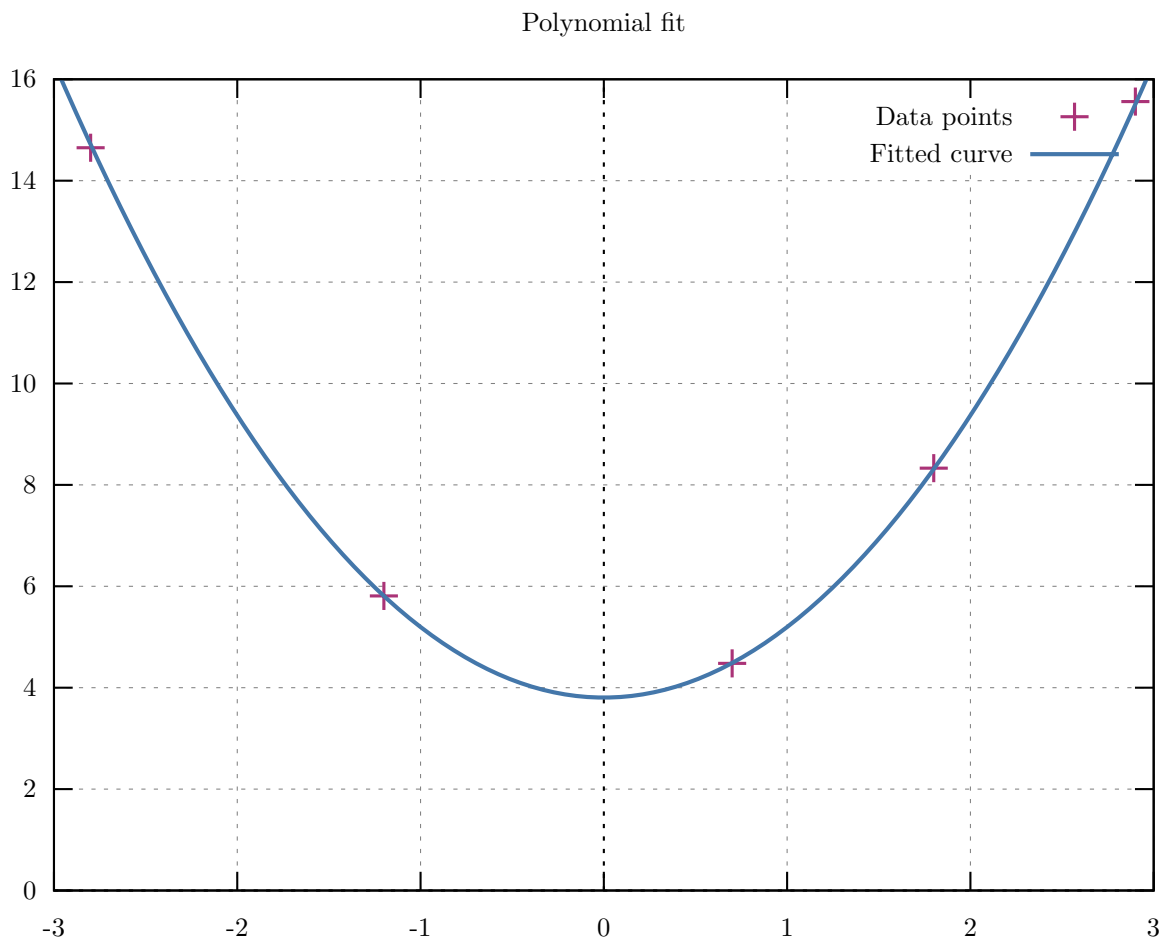


Figure 13: Polynomial fit for Problem Set 2 (vii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 1.39137	+/- 0.006722	(0.4831%)
b	= 3.80539	+/- 0.0362	(0.9513%)

Question 7

Problem Set 2 (ix). Fit the data with $f(x) = x^a$ and report a . Plot the data with the fitted function, title the plot *Data fitting*, and label the axes x and $f(x)$.

Table 7: Data for Problem Set 2 (ix).

x	$f(x)$
0.0	0.0
0.2	0.006
0.4	0.053
0.6	0.195
0.8	0.49
1.0	1.0
1.2	1.792
1.4	2.935
1.6	4.5
1.8	6.56
2.0	9.19

Data file contents:

```
0.0 0.0
0.2 0.006
0.4 0.053
0.6 0.195
0.8 0.49
1.0 1.0
1.2 1.792
1.4 2.935
1.6 4.5
1.8 6.56
2.0 9.19
```

Sol: The fitted plot and parameter estimates with their standard errors are shown below.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_ix_ou.svg"
4
5
6 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_ix.log"
7
8 set grid
9 set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11
12 set title "Data fitting"
13 set xlabel "x"
14 set ylabel "f(x)"
15
16 y(x)=x**a
17
18 fit y(x) "Lab-files/Data/2/Q_ix.txt" using 1:2 via a
19 plot "Lab-files/Data/2/Q_ix.txt" using 1:2 with p ps 1.5 lc "#228833" title "Data
    points", \
20     y(x) lw 2 lc "#AA3377" title "Fitted curve"
```

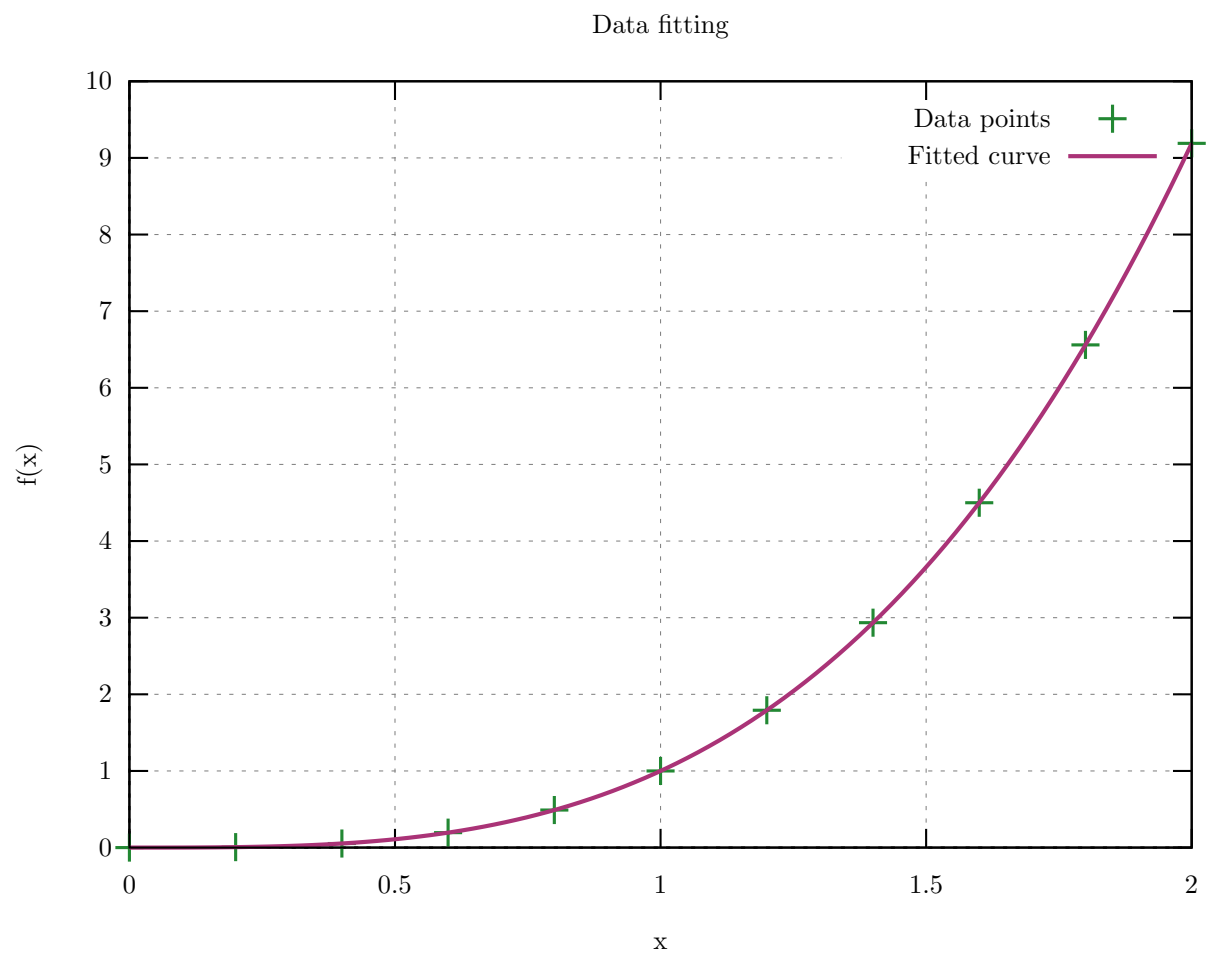


Figure 14: Power fit for Problem Set 2 (ix).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 3.20008	+/- 2.388e-05	(0.0007463%)

Section 3

Question 1

Problem Set 3 (i). Show the polar plots of $\sin^3(t)$ and $\cos^2(2t)$. Label the axes “t value” and “the range”, place “polar plot” at $(0, -0.7)$, and show the vertical axis. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_i_ou.svg"
6
7  set title "Polar plot"
8  set grid polar linestyle 0 lc "black"
9  set xlabel "t value"
10 set ylabel "the range"
11 set label "polar plot" at 0,-0.7
12 set yzeroaxis linestyle 1 lc "black"
13
14 set samples 1000
15 plot (sin(t))**3 lw 2 lc "#4477AA",cos(2*t)**2 lw 2 lc "#EE7733"
```

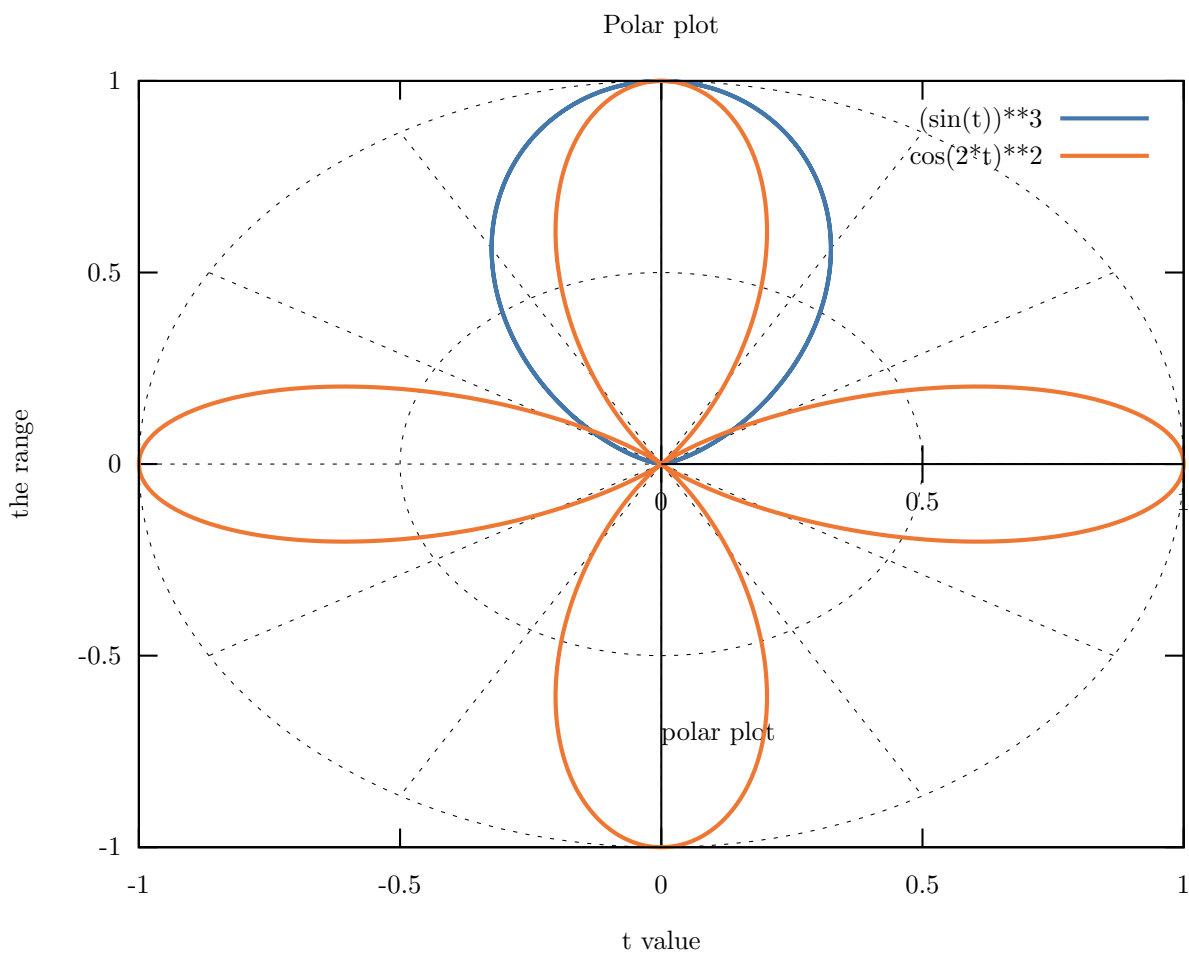


Figure 15: Polar plots of $\sin^3(t)$ and $\cos^2(2t)$.

Question 2

Problem Set 3 (ii). Show the polar plots of $\tan(6t)$ and $\cot(6t)$ over ranges -20 to 20 on both axes. Place the label “ $\tan(6t)$, $\cot(6t)$ ” at $(4, 12)$ and show both axes. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_ii_ou.svg"
6
7  set title "Polar plot"
8
9  set grid polar linestyle 1
10 set xlabel "t value"
11 set ylabel "the range"
12 set xzeroaxis linestyle 1 lc "black"
13 set yzeroaxis linestyle 1 lc "black"
14 set label "tan(6t), cot(6t)" at 4,12
15
16 set xrange [-20:20]
17 set yrange [-20:20]
18
19
20 set samples 1000
21
22 plot tan(6*t) lc "#332288", (1/tan(6*t)) lc "#CC6677"
```

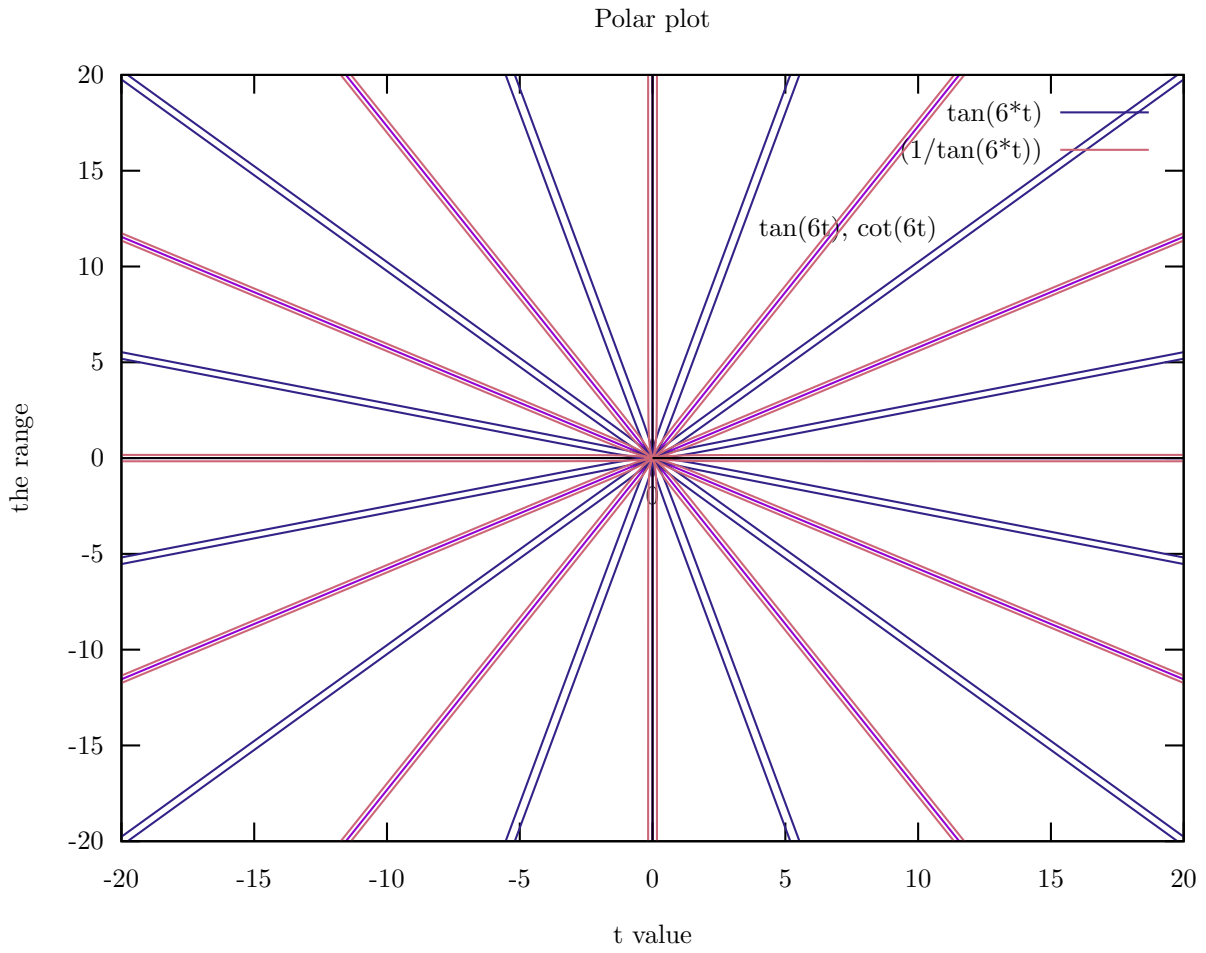


Figure 16: Polar plots of $\tan(6t)$ and $\cot(6t)$.

Question 3

Problem Set 3 (iii). Plot the parametric curve $x = 2 \cos^3 t$, $y = 2 \sin^3 t$ for $-\pi \leq t \leq \pi$. Label the axes as specified and title the plot "Astroid". **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set parametric
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_iii_ou.svg"
6
7  set title "Astroid"
8  set xlabel "x = 2 cos(t)^3"
9  set ylabel "y = 2 sin(t)^3"
10 set grid linestyle 1 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [-pi:pi]
16 x(t)=2*cos(t)**3
17 y(t)=2*sin(t)**3
18
19 set samples 1000
20
21 plot x(t),y(t) lw 2 lc "#228833"
```

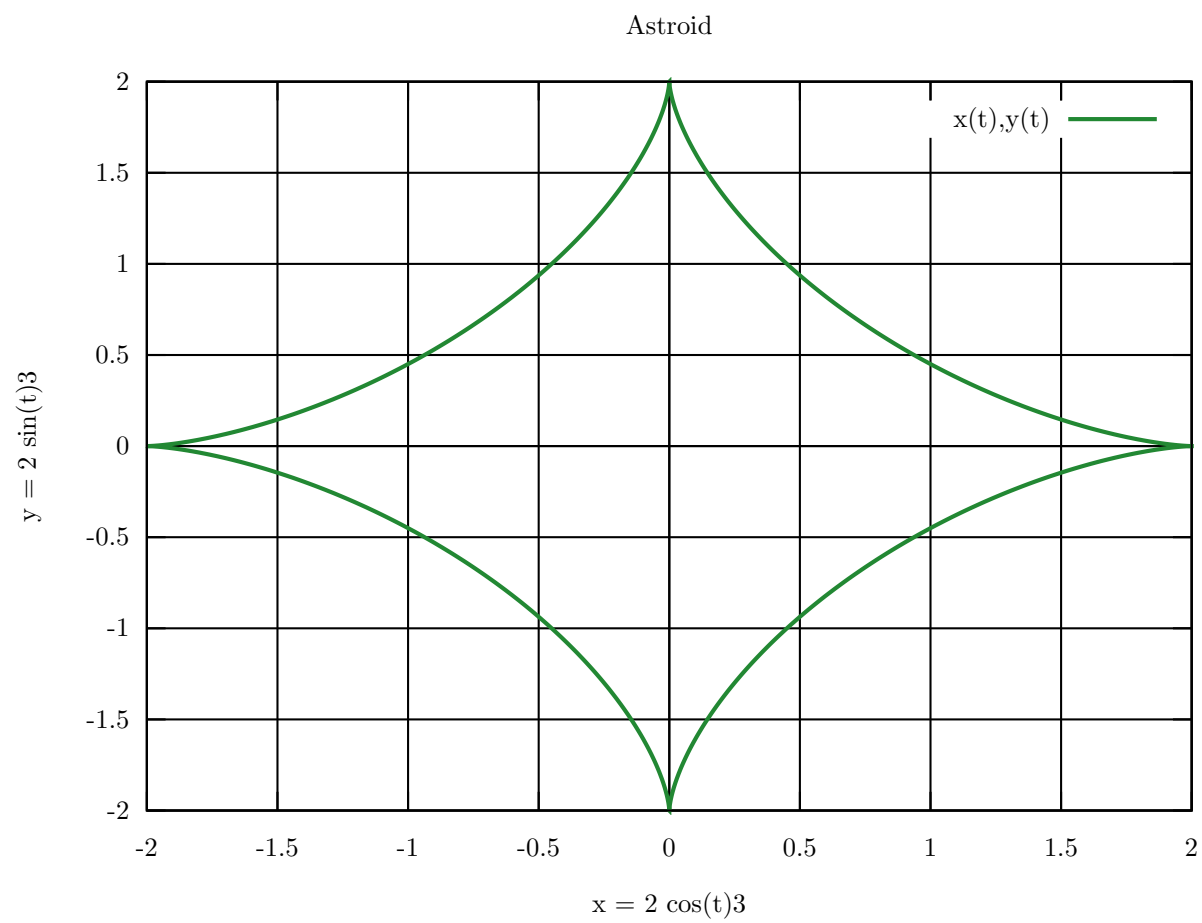


Figure 17: Astroid parametric plot.

Question 4

Problem Set 3 (iv). Plot the polar spiral $r = t$ for $0 \leq t \leq 6\pi$ using 10000 samples. Show both axes, label them as specified, and title the plot “Spiral”. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_iv_ou.svg"
6
7  set title "Spiral"
8  set xlabel "x = r cos t"
9  set ylabel "y = r sin t"
10 set grid polar lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [0:6*pi]
16 r(t)=t
17
18 set samples 10000
19
20 plot r(t) lw 2 lc "#AA3377"
```

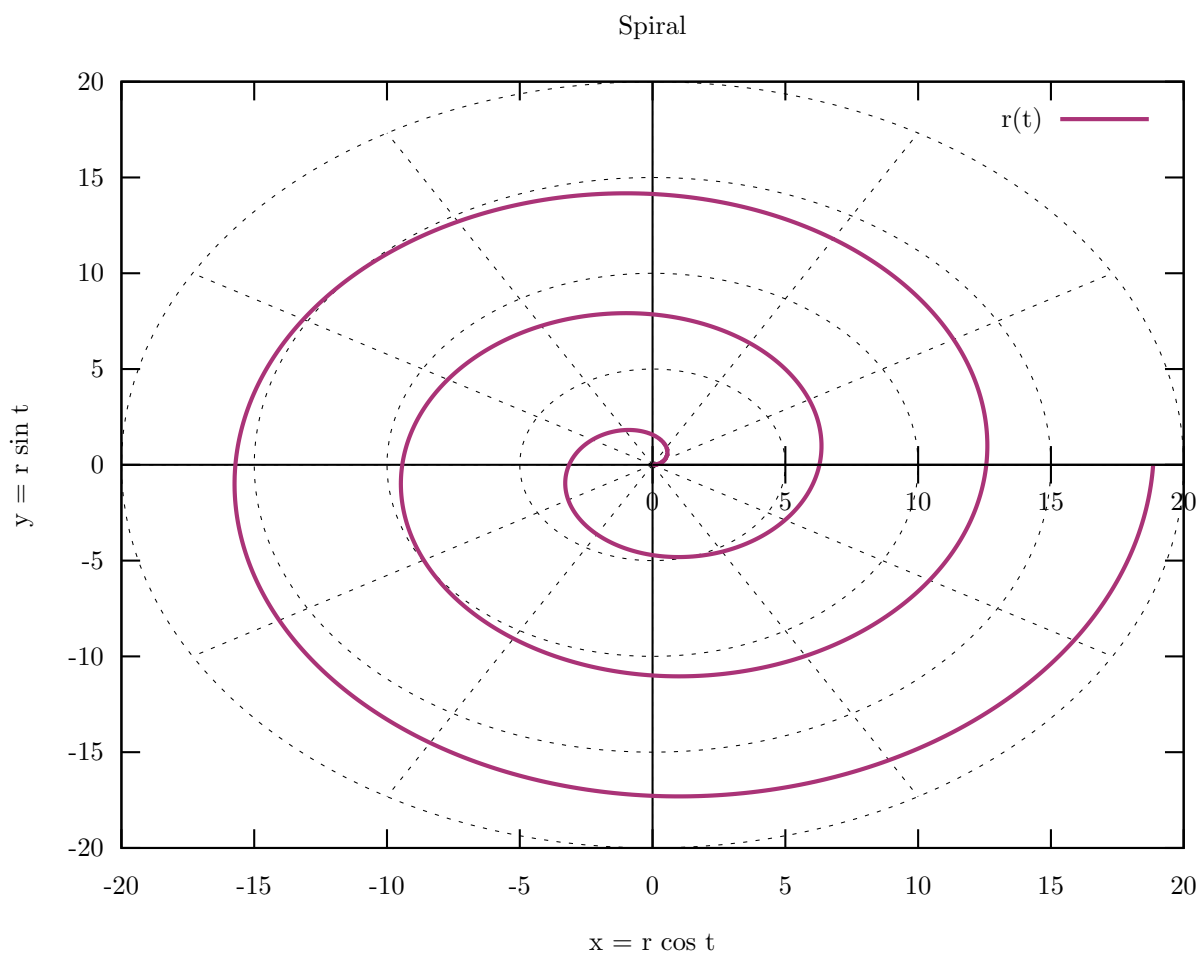


Figure 18: Polar spiral $r = t$.

Question 5

Problem Set 3 (v). Plot the parametric curve $x = 2 \sin t$, $y = 2 \sin(2t)$ for $-\pi \leq t \leq \pi$, with the specified axis labels and title “Lissajous Figure”. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set parametric
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_v_ou.svg"
6
7  set title "Lissajous Figure"
8  set xlabel "x = 2 sin t"
9  set ylabel "y = 2 sin 2t"
10 set grid linestyle 0 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [-pi:pi]
16
17 x(t)=2*sin(t)
18 y(t)=2*sin(2*t)
19
20
21 set samples 1000
22
23 plot x(t),y(t) lw 2 lc "#EE7733"
```

Lissajous Figure

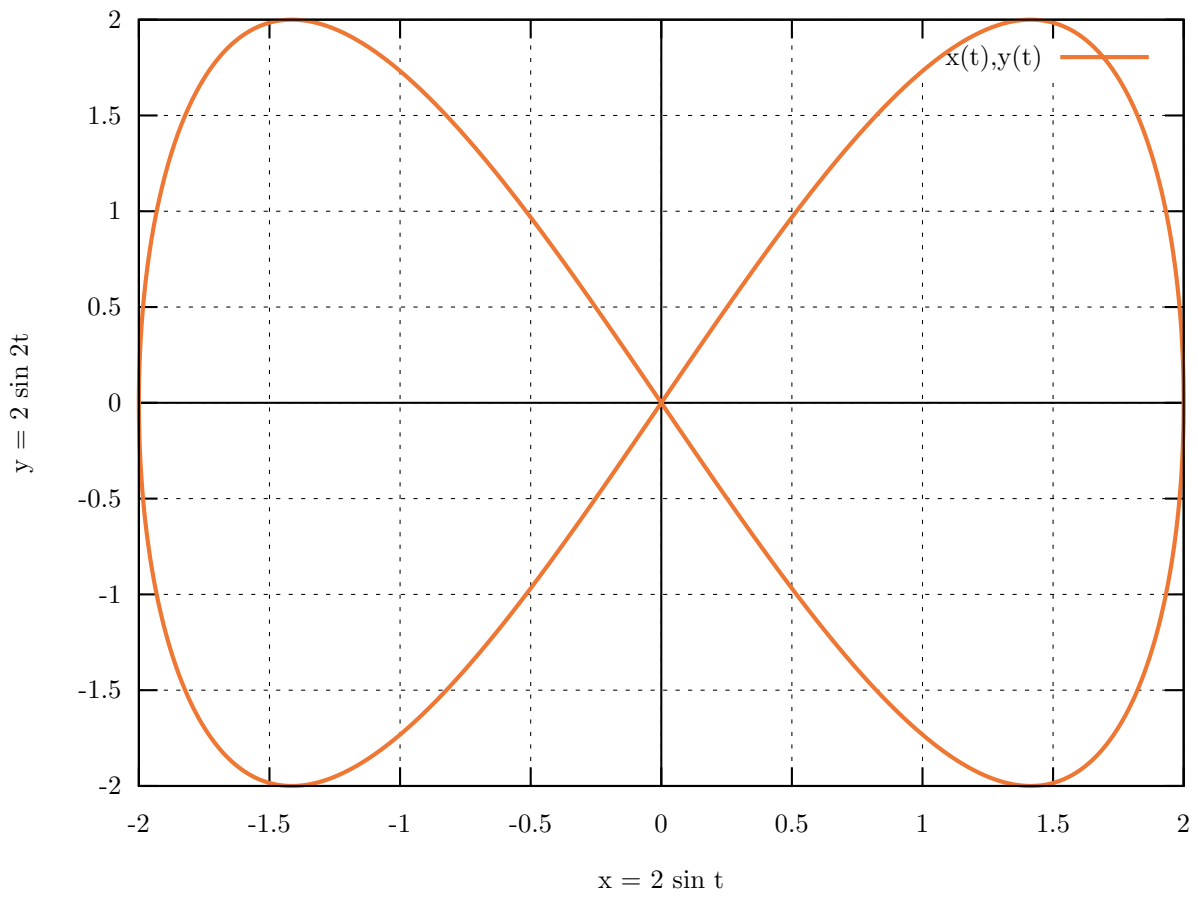


Figure 19: Lissajous parametric plot.

Question 6

Problem Set 3 (vi). Plot the polar curves $r_1 = 4 + 3 \cos t$ and $r_2 = 3 + 4 \cos t$. Label the axes as specified, place “polar plot” at $(-6, 6)$, and title the plot “A pair of Limacon”. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_vi_ou.svg"
6
7  set title "A pair of Limacon"
8  set xlabel "x = r cos t"
9  set ylabel "y = r sin t"
10 set grid linestyle 0 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13 set label "polar plot" at -6,6
14
15 set trange [-pi:pi]
16
17 r1(t)=4+3*cos(t)
18 r2(t)=3+4*cos(t)
19
20
21 set samples 1000
22
23 plot r1(t) lw 2 lc "#4477AA", r2(t) lw 2 lc "#AA3377"
```

A pair of Limacon

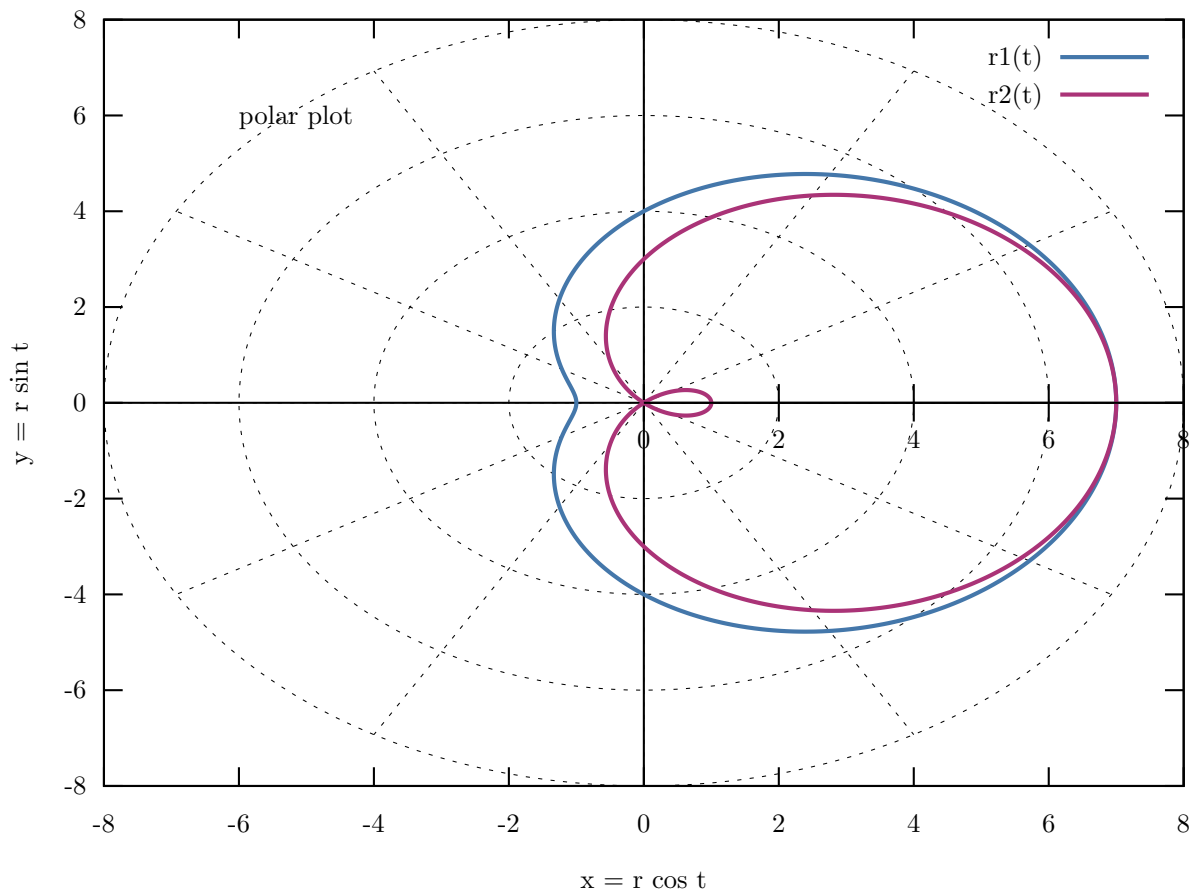


Figure 20: A pair of limaçons.